Resource Allocation for Semantic Aware Relay Networks Using Multi-Agent Reinforcement Learning

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***Abstract*—The variety and volume of real-time video streaming services make network resources like power, frequency, and time critical to the user experience. Semantic communication, by transmitting only meaningful data, significantly reduces the load on the network, leading to faster transmission times and reduced latency—both crucial for enhancing user experience. In this paper, we develop a context-aware resource allocation framework for a semantic-aware regenerative AI-based Unmanned Aerial Vehicle (UAV) setup that accommodates both conventional and semantic communication users. Specifically, we define a semantic feature pooling mechanism, upon which a novel Quality of Experience (QoE) model is proposed. For dynamic network environments, we formulate a long-term resource allocation problem by maximizing the expected rewards. Each UAV is modeled as a learning agent, with each resource allocation solution corresponding to an action taken by the UAVs. We then develop a Multi-Agent Reinforcement Learning (MARL) framework in which each agent discovers its optimal strategy based on local observations. More specifically, we propose an agent-independent method where all agents execute a decision algorithm independently while sharing a common structure based on Q-learning. The proposed framework also facilitates intelligent traffic steering by dynamically adjusting resource distribution based on real-time context. Finally, simulation results demonstrate the effectiveness and superiority of the proposed method in terms of overall QoE.**

***Index Terms*—Multi-agent reinforcement learning, resource allocation, semantic communication.**

I. Introduction

The exponential growth of wireless data traffic is pushing communication networks to gain high efficiency and intelligence. Efficient handling of real-time ultra-large-scale connections to enable smart applications, considering limited network resources, introduces significant challenges for the development of 6th-generation Radio Access Network (RAN) systems. The existing communication paradigm has focused on accurate symbol transmissions, but due to high-volume traffic applications (e.g., YouTube, TikTok, Facebook, etc.), network resource scarcity is touching its limits. As a promising alternative, the semantic communication paradigm has attracted research attention, as it utilises the knowledge of the source to transmit only semantic information at the transmitter side and recover it on the receiving side [1]. To further improve efficiency, context-aware mechanisms can help adapt communication strategies based on situational network and user information, enabling more intelligent and proactive traffic steering to balance loads and optimize performance. [1].

On the other hand Aerial communication systems, encourage new innovative functions to deploy wireless infrastructure. Such as rapid deployment for complementing terrestrial communications based on the 3rd Generation Partnership Project (3GPP) Long Term Evolution-Advanced (LTE-A) [2]. The major advantage is in contrast to channel characteristics of terrestrial communications, the channels of UAV-to-ground communications are more probably Line-of-Sight (LoS) links [3], which is beneficial for wireless communications. In particular, UAVs based on different aerial platforms that provide wireless services have attracted extensive research and industry efforts for the issues of deployment, navigation, and control. However, resource management issues remain the top priority. Thankfully, semantic communication and regenerative Artificial Intelligence (AI) can significantly reduce the transmission resource burden. Because, regenerative AI utilises semantic representations extracted by semantic communication for guiding the desired user content. Many research works focused on UAV semantic communications but ignored the resource allocation aspect. Furthermore, conventional resource allocation methods are inadequate for semantic communications as additional semantic information is also involved in the resource allocation process. For existing semantic communication-aware resource allocation methods, only focused on non-realtime applications with simple network settings assumptions (e.g., single cell, single hop, and a network with semantic communication users only).

Therefore, in this work, we first design semantic communication-based QoE model for live streaming applications and then develop MARL based resource allocation scheme for regenerative AI enhanced UAV relay networks.

A. Prior Works

In existing research efforts, various works are proposed as end to end black box based designs based on Deep Neural Networks (DNNs) [4]. Which is extended by the authors in [5] to transmit text and speech data streams. Similar, works were further extended to multi-modal semantic communication designs. Along with these architectural designs, some works also focus on resource allocation and very few on semantic communication-based QoE enhancements. For example, authors in [6] proposed QoE enhancements by optimizing transmitted symbols and channel assignments. However, the majority of existing works are based on DNNs, which brings functional limitations because it is handled as a black box. Additionally, training data availability for real-time applications is hard to obtain.

Similarly, very few works also focused on semantic relay transmissions, as in [5–7] representative works. Among these works, [5, 7] are based on black box-based end-to-end design while focusing only on text transmissions which are not resource demanding as volumetric video-like traffic. In addition, no existing work is focused on enhancing semantic communications for real-time applications [8].

To counter these problems, we first propose a regenerative relay design for semantic RAN to support conventional as well as semantic communication users. The relay can dynamically adjust to the network traffic settings and regenerate the live-streaming contents for resource constrained users while transmits the regenerated streaming content features to users which are capable to process semantic features locally. Second, we built on this design and proposed a novel QoE model with a smart multi-agent resource allocation scheme to enrich the network resource utility further.

II. System Model And Problem Formulation

This paper investigates UAV-aided relays for downlink semantic communications with real-time video transmissions, as shown in Fig. 1. In our model, two types of remote users, i.e., semantic and conventional communication users, are connected to central BS via sub-channels $\mathcal{S} = \{1, 2, \cdots, S\}$ for live video streaming. The set of users $\mathcal{U} = \{1, 2, \cdots, U\}$ represents semantic communication users, whereas conventional communication users are denoted by $\mathcal{I} = \{1, 2, \cdots, I\}$. Semantic communication users receive only the semantic features from UAVs necessary for regenerating the content, rather than the entire raw data (such as object positions, gestures, or key points, which can be used to reconstruct the content at the receiver's end). In contrast, conventional communication users depend on frame-by-frame transmission, where every frame of video or image data is sent. We assume that the direct link between BS and users is blocked (e.g, high walls around a stadium environment, etc). Therefore, $\mathcal{V} = \{1, 2, \cdots, V\}$ UAV relays are deployed between devices and BS for effective content delivery. Furthermore, we assume that the number of UAV relay nodes is less than the number of active users in the network $V < \hat{U}$; likewise, a single UAV can serve multiple users. Additionally, users can only associate with one UAV at a time. Finally, one transmission cycle requires two time periods to complete. Thus, users will be able to receive the information in the second time slot.

A. Transmission Model

The distances of BS to UAV (v) and UAV (v) to the user (u) respectively $d_{o,v}$ and $d_{u,v}$, where the relay is placed at height H. Similarly, $\Delta d_{v_1,v_2}$ is the separation distance between UAV 1 and UAV 2 should be greater than minimum separation distance d_{min}. Based on this setup, we calculate the channel gains for the BS to UAV (i.e., $h_{o,v}$) and UAV to users (i.e., $h_{u,v}$), respectively, as follows: [3]:

$$h_{o,v} = \frac{\beta}{[d_{o,v}]^{\alpha/2}}, \tag{1}$$

$$h_{u,v} = \frac{\beta}{[d_{u,v}]^{\alpha/2}}, \tag{2}$$

where α is the path loss exponent and β is the channel power gain when the reference distance is 1 meter distance. The transmission rate based on the receiving signal-to-noise ratio (SINR/SNR) for the UAV v at subchannel s is defined in (3) and for downlink transmissions, the throughput for the UAV-based relay is calculated using the function (4) defined as follows [9, 10]:

$$R_{o,v} = B \log_2(1 + \frac{p_{o,v} h_{o,v}}{N_0}), \tag{3}$$

where $p_{o,v}$ is the received power of the $v-th$ UAV from BS with channel gain $h_{o,v}$ and N_0 is Additive white Gaussian noise (AWGN). Accordingly, the throughput for UAV (v) can be expressed as

$$T_{o,v} = \sum_{v=1}^{V} R_{o,v}, \tag{4}$$

regarding users, we denote that the direct link between users and BS is blocked. Therefore, all the users can connect via relay nodes. The transmission rate based on the receiving signal-to-interference-plus-noise ratio (SINR) of users from relay nodes is defined in the following (5), followed by the throughput calculation function (6).

$$R^s_{u,v} = B \log_2(1 + \frac{p_{u,v} h_{u,v} c^s_v}{\sum\limits_{i \in V \setminus v} c^s_{u,i} p_{u,i} h_{u,i} + N_0}) \tag{5}$$

where $p_{u,v}$ is the received power of the $u-th$ user from $v-th$ UAV with channel gain $h_{u,v}$, $\sum\limits_{i \in V \setminus v} c^s_{u,i} p_{u,i} h_{u,i}$ is inter-UAV interference and N_0 is AWGN.

$$T^s_{u,v} = \sum_{u=1}^{U} R^s_{u,v} \tag{6}$$

similarly, the association for user u/i, sub-channel s, and serving UAV v is expressed as

$$c^s_{u,v} = \begin{cases} 1, & \textit{if any user } u/i \textit{ is associated to UAV } v \\ & \textit{via sub-channel } s, \\ 0, & \text{otherwise.} \end{cases} \tag{7}$$

B. Semantic Features and QoE Modelling

We assume a BS has multiple live-streaming contents $\mathcal{C} = \{1, 2, \cdots, C\}$ to deliver to multiple users via UAVs. Therefore, content c is transformed into $Z_n(t)$ features containing $\mathbb{Z}^6$, representing the x-axis, y-axis, z-axis, and RGB colour channels. Based on $Z_n(t)$ features, we define a set of feature pools

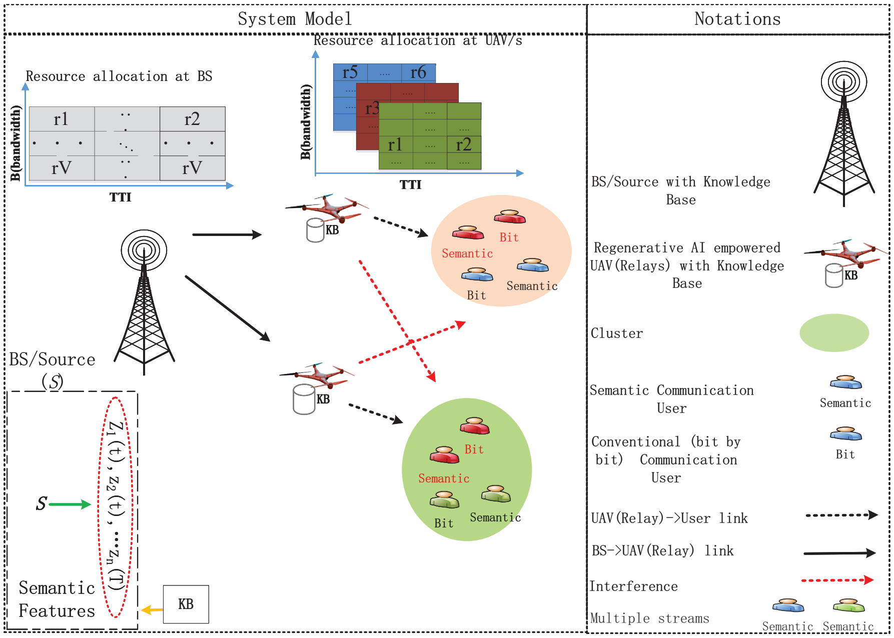

Fig. 1: Illustrating the proposed system model: for downlink transmissions, multiple UAVs are simultaneously serving semantic communication and conventional users.

as $\hat{\mathbf{P}} = \{\hat{\mathbf{p_1}}, \hat{\mathbf{p_2}}, \cdots, \hat{\mathbf{p_L}}\}$ and QoE score as $Q(\vec{n})$, where $\vec{n} = \langle \omega n_{\hat{p_1}}, \omega n_{\hat{p_2}}, \cdots, \omega n_{\hat{p_L}} \rangle$ contains QoE score $\omega n_{\hat{p_\ell}}$ with importance weight ($\omega \in [0-1]$) for each feature pool $\hat{\mathbf{p}}_\ell$. A higher score means more importance and vice versa. Similarly, the QoE score for UAV (v), semantic communication user (u), and bit-by-bit communication user (i) can be defined as $Q(\vec{n_v})$, $Q(\vec{n_i})$, and $Q(\vec{n_u})$.

C. Problem Formulation

In order to derive an optimal resource allocation strategy in this part, we formulate resource allocation for downlink multi-user multi-relay semantic communications. Based on the introduced quantities described above, we can construct the average QoE maximisation problem for the proposed downlink semantic communication for relay based system. The optimization problem can be expressed as follows:

$$\max_{\pi, c^s_{u,v}} E\left(\sum_{v=1}^{V} Q(\vec{n_v}) + \sum_{i=1}^{I} Q(\vec{n_i}) + \sum_{u=1}^{U} Q(\vec{n_u})\right), \tag{8a}$$

$$s.t.$$

$$C_1 : c^s_{u,v} \in \{0,1\}, \forall s, u, v \tag{8b}$$

$$C_2 : p_{u,v} \geq 0, \forall u, v, \tag{8c}$$

$$C_3 : p_{u,v} \leq P_{tot,u}, \forall s, u \tag{8d}$$

$$C_4 : p_{o,v} \geq 0, \forall v \tag{8e}$$

$$C_5 : p_{o,v} \leq P_{tot,i}, \forall s, i, \tag{8f}$$

$$C_6 : T_{o,v} \geq T_{min}, \forall v, s, \tag{8g}$$

$$C_7 : T^s_{u,v} \geq T_{min}, \forall s, u, i, \tag{8h}$$

$$C_8 : d_v \in \mathcal{D}, \forall v, \tag{8i}$$

$$C_9 : \lambda_v(\%) \in [0-100], \tag{8j}$$

Where constraint C_1 indicates the resource allocation variable can be 0 or 1. Constraints $C_2 - C_5$ are power allocation constraints for different network users and UAVs, showing that the allocated is greater than 0 and less than total power. Similarly, constraints C_6 and C_7 are throughput constraints for the users and UAVs, indicating that the minimum tolerable throughput threshold is greater than T_{min}. Constraint C_8 considers that the positions of UAVs are subject to minimum collision avoidance distance d^2_{min} in the considered region $\mathcal{D} = \{\mathbf{d^r} \in \mathbb{R}^3 | [x^r]_1 \in [0, x_{\text{D}}], [y^r]_2 \in [0, y_{\text{D}}], [z^r]_3 \geq h_{\text{min}}\}$. C_9 shows the active regenerative traffic volume at each UAV. The problem in (8) is classed as an NP-hard problem, even if only constraint C_3 is considered [11]. To handle such types of problems, Learning-based strategies perform better than conventional methods to handle such types of problem section, we transform this problem to solve using advanced ML algorithms.

III. Markovian Game Framework for Multi-UAV Semantic RAN

This section explains the proposed MARL framework for Multi-UAV Semantic RAN. Then we propose the Q-learning based resource assignment algorithm for maximizing the long-term expected reward of the considered multi-UAV based semantic RAN.

A. Problem Reformulation and Design

In this subsection, we consider modeling the formulated problem (8) by adopting a stochastic game (also called Markov game) framework [11], since it is the generalization of the Markov decision processes to the multi-agent case. For MARL, the design of the Markov Decision Process (MDP) is vital. In this work, the basic MDP consists of an agent/s (UAVs) that interact with the wireless environment E by performing actions a_m to change its current s_m to future state s'. For each action, a reward r is offered to the agent to reward better performance or punish worse performance when compared with the previous state. The main goal is to reach the final state and action pair, which contains the optimal (sub-optimal) objectives.

An MDP consists of a tuple of $\big(\mathbf{N}, \mathbf{S}, \mathbf{A}, \mathbf{R}(.), \mathbf{E}$ and $f(.)\big)$, where $\mathbf{N}$ is the number of agent(s) (BS, GF clients), $\mathbf{S}$ is the set of states, actions are denoted by $\mathbf{A}$, and $\mathbf{R}(.)$ is the reward function.

Definition 1. *Let a finite set of states $\mathcal{S} = \{s_1, \cdots, \mathbf{S}\}$ and $\mathbf{S}\times\mathbf{S}$ transition matrix $\mathbf{F}$ with $0 \leq f \leq 1$ for any s. The process starts with the second action after successfully completing the first one. So, the probability of moving to the next state is as follows:*

$$\Pr\{s(t+1) = s_j | s(t) = s_j\} = f, \tag{9}$$

To start the learning process, RL agents interact with the environment to maximize the long-term reward following some policy π.

- **Agent(s):** Agents in our case are UAVs denoted by $\mathcal{M} = \{1, \cdots, M\}$. MDP for the MARL agent/s (UAVs) m depends on the following elements.
- **State:** Since each user belongs to only one sub-channel/cluster according to (7), $\mathbf{S}$ is a finite set consisting of $(S)^U$ states. For UAV m every state s_m represents one combination of 2D associations among users and sub-channels.
- **Actions (UAVs):** $\mathbf{A}$ is an action space consisting of three actions $[-1, 0, +1]$. For each sub-channel/cluster, the '-1' is to move out one user to be an unconnected user, the '$+1$' is to accept the unconnected user, and the '0' represents no change in this sub-channel. To keep the users connected while ensuring the number of users is the same, we take two actions at the same time as a swap operation.
- **Rewards:** In MDP, the agent tries to maximize the sum reward for each learning time-step (t) based on the transition of current state action pair $\{s, a\}$ to the next state and action pair $\{s', a'\}$. To perform reliable resource allocations, we use the semantic QoE score as a dynamic reward function at each time-step (t). The dynamic reward function can be expressed as follows:

$$R = \begin{cases} E\Big(\sum\limits_{v=1}^{V} Q(\vec{n_v}) + \sum\limits_{i=1}^{I} Q(\vec{n_i}) + \sum\limits_{u=1}^{U} Q(\vec{n_u})\Big), & \text{(8a)}(t) \geq \\ & \text{(8a)}(t-1) \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

  and the long-term expectation is,
  $R_m(s_m, a_m, s'_m) = E\{r'_m | s^t_m = s_m, a^t_m = a_m, s'_m = s'_m\}$.
- **Environment:** Environment in our case is a semantic RAN environment that consists of multiple users (i.e., semantic and conventional communication users), central BS, and UAVs (relays/Agents).
- **Transition Probability function:** The state and action transition function shows the state transition probabilities of the agent for changing its current state to the next in search of the optimal (sub-optimal) state and action pair. The transition function for the proposed method is given by $f_{s_m \to s'_m} = f(s_m, a_m, s'_m)$.

For each UAV (agents), we define a Q-function as the expected cumulative discounted reward to find the optimal policy π^*, which can be given as follows:

$$Q_m(s_m, a_m, \pi) = E\left\{\sum_{\tau=0}^{+\infty} \delta^\tau r_m^{t+\tau+1} \mid s^t = s, a^t_m = a_m\right\}, \tag{11}$$

where r_m is the discounted reward. Similarly, the state value function is defined as follows:

$$\begin{aligned} V_m(s_m, \pi) =& E\left\{\sum_{\tau=0}^{+\infty} \delta^\tau r_m^{t+\tau+1} | s^t_m = s_m\right\} \\ &= \sum_{s'_m \in \mathcal{S}_m} f(s_m, a, s'_m) \sum_{a \in \mathcal{A}} \prod_{n \in \mathcal{M}} \pi_n(s_n, a_n) \\ &\quad \times \left[R_m(s_m, a, s'_m) + \delta V(s'_m, \pi)\right], \end{aligned} \tag{12}$$

where $\pi_n(s_n, a_n)$ shows the action a_n selection probabilities for state s_n for agent m. Similarly, based on (12) the Q function can be written in recursive form as

$$\begin{aligned} &Q_m(s_m, a_m, \pi) \\ &= E\left\{r_m^{t+1} + \delta \sum_{\tau=0}^{+\infty} \delta^\tau r_m^{t+\tau+2} | s^t_m = s_m, a^t_m = a, s^{t+1}_m = s'_m\right\} \\ &= \sum_{s'_m \in \mathcal{S}_m} F(s_m, a, s'_m) \sum_{a_{-m} \in \mathcal{A}} \prod_{j \in \mathcal{M} \setminus \{m\}} \pi_j(s_j, a_j) \\ &\quad \times \left[R(s_m, a, s'_m) + \delta V_m(s'_m, \pi)\right]. \end{aligned} \tag{13}$$

from (12) and (13) the following relationship can be derived between Q-values and state values:

$$V_m(s_m, \pi) = \sum_{a_m \in \mathcal{A}} \pi_m(s_m, a_m) Q_m(s_m, a_m, \pi). \tag{14}$$

The main objective of MARL agents is to reach the optimal resource allocation strategy by reaching maximal reward. The value function for the optimal policy can be defined as follows:

$$V^*_m = \max_{\pi_m} V_m(s_m, \pi), \quad s_m \in \mathcal{S}. \tag{15}$$

Remark 1. *We remark that each agent UAV have partial knowledge of the underline wireless network environment via the obtained rewards of the other agents.*

Remark 2. *The convergence rate of the proposed algorithm varies according to the initial 3D association (states) that is randomly selected. In this model, the state space means allocation strategies that include subsets of all possible associations of active users for each UAV at the Long-term timeslot T_e.*

TABLE I: Network settings

Time-slots	500
Time-steps	500
Total BSs	1
Total UAVs	2
Total users	2
Total resource blocks	3
Optimisers	Q-Table
Dynamic-Reward	Data-rate (t)
Bandwidth	1 MHz
Path loss	4
Traffic Types	hybrid (semantic+conventional)
σ^2	-174 dBm
γ	0.6
α	0.75
$\epsilon - greedy$	0.01
λ_v **(%)**	[0-100]

B. The Algorithm Details

Based on the above discussions, we design **Algorithm 1** for step by step significant optimization stages of the MARL base resource allocation algorithm. The details of the mentioned algorithm are as follows:

- Input/s and Initialization: we initialize the network and learning parameters before the start of the agent/s process. All agents (UAVs) use a ϵ-greedy policy to explore the environment. The agents receive states from the environment, and they take action. Based on the action taken, agents obtain a reward and the next state from the environment.
- Key steps for Learning: From Line number 7 till the end, show the main learning process for all agents (UAVs). Each agent performs actions based on the current state of the wireless network environment, and user sub-channel associations. Suppose the agent successfully picks the batter association that results in a reward (QoE score) and any agent fails. In that case, the agent receives a punishment of 0 to enhance the local as well as global rewards. All of these interactions are performed, and then each agent updates its Q-Table. This whole process continues until all the agents can reach the best possible outcomes in the form of optimal rewards.

Complexity Analysis: For each agent, the complexity of the proposed framework is mainly decided by the number of communicating users and the total number of sub-channels. More specifically, the computation complexity for our framework is based on the trace-table $\mathcal{O}(Q_m(s_m, a_m))$, the total number of states $(S)^U$, and actions in $\mathbf{A}$. Therefore, the overall complexity can be denoted as $\mathcal{O}(Q_m(s_m, a_m))$.

IV. Simulation Results

In this section, we verify the effectiveness of the proposed MARL-based resource allocation algorithm for multi-UAV networks by simulations. The simulation parameters and system settings for the proposed model is presented in TABLE I. An Intel Core i7-12700H 3.6 GHz CPU desktop with 16 GB

Algorithm 1 Dynamic Resource Allocation Framework for Multi-UAV based Downlink Semantic Communications

1: Inputs for MARL:
   1) Episodes M
   2) Episodes T_e
   3) Explorations per trials T_t
   4) Learning rate α

2: Initialization for MARL:
   1) Network parameters (U, V, S)
   2) Q-Tables $Q_m(s_m, a_m)$

3: Define the number of clusters
4: Define the range of users in the network
5: load $\mathbf{s} = \mathbf{S^U}$ and $\mathbf{a} = [-1, 0, +1]$
6: Random user association to any sub-channel at each UAV
7: **for** iteration = 1:T_e **do**
8: $s_t = rand()$
9: **for** iteration = 1:T_t **do**
10: **for** Agent = 1:M **do**
11: s_t, a_t
12: compute $R = \begin{cases} E\left(\sum_{v=1}^{V} Q(\overrightarrow{n_v}) + \sum_{i=1}^{I} Q(\overrightarrow{n_i}) + \sum_{u=1}^{U} Q(\overrightarrow{n_u})\right), & \text{(8a)}(t) \geq \text{(8a)}(t-1) \\ 0, & \text{otherwise.} \end{cases}$
13: update R
14: update $Q_m^*(s_m, a_m) = \sum_{a_{-m} \in A_{-m}} \prod_{j \in \mathcal{M} \setminus \{m\}} \pi_j(s_j, a_j) \times \sum_{s'_m} F(s_m, a, s'_m) \left[R(s_m, a_m, s'_m) + \delta \max_{a'_m} Q_m^*(s'_m, a'_m)\right].$
15: Update π towards greediness
16: $s \leftarrow s', a \leftarrow a'$
17: return optimised $(c_{b,t})$ (8a) under constraints and (8)

random access memory is used for all the simulations. The deployment and parameters setup (TABLE I) of the multi-UAV network are mainly based on the investigations in [3, 12].

Fig. 2 shows the deployment of the BS, UAVs, and users $50m\ x\ 50m$ grid area. Where the BS is placed at a $(0, 0)$ location, users are randomly distributed in two clusters around the centroid of each cluster. The UAVs are placed in the middle of the centroid of each cluster and the BS. For illustrative purposes, Fig. 3 shows the average reward and the average reward per time slot of the UAVs under the setup of Fig. 2. As it can be observed, the average reward increases with the increase in algorithmic episodes. Interestingly, we can also see that the proposed algorithm archives convergence in a short time. That shows the long-term adaptability of our algorithm in a very dynamically changing environment. In Fig. 4, we investigate the QoE performance of conventional communication with round-robin resource allocations, Semantic communication with round-robin resource allocation, and MARL-based semantic communication strategies. This figure is revealing in several ways. Firstly, semantic communication is performing better than conventional communication. Secondly, the proposed MARL-based intelligent resource allocation is more robust to changing network loads than conventional resource allocation strategies. Therefore, we can see the long-term benefits of the proposed MARL-based resource allocation and semantic communication as compared with conventional communication and round-robin based resource allocation strategy. Lastly, long-term average QoE is inversely proportional to the number of network users. Due to this reason, an increase in network users results in decreasing QoE scores. It is also interesting to mention that the performance of MARL-based

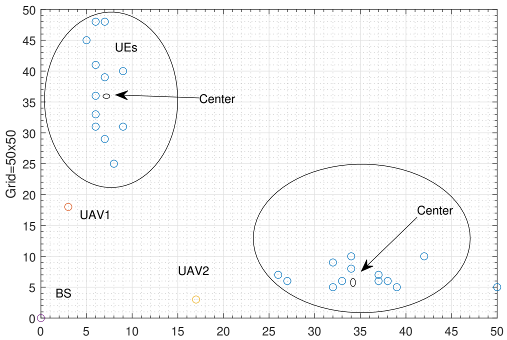

Fig. 2: Illustration for BS, UAV, and user deployment.

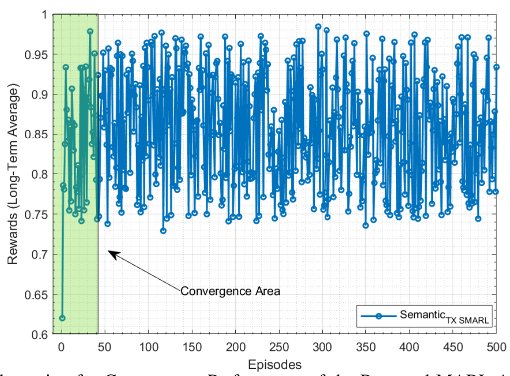

Fig. 3: Illustration for Convergence Performance of the Proposed MARL Algorithm.

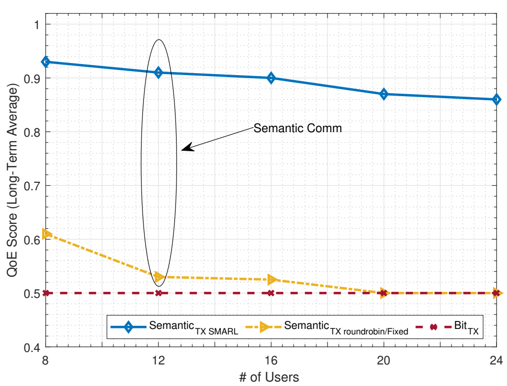

Fig. 4: Illustration for Performance of long-term Average QoE with respect to the number of semantic users, bit transmission oriented users, Round Robin and MARL based resource allocation schemes, when $\lambda_v = 50\%$

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semantic communication is better than other schemes while supporting $\lambda_v = 50\%$ conventional traffic. This proves the dynamicity and robustness of MARL based semantic communication algorithm.

V. Conclusion

In this paper, we proposed a novel QoE design for semantic communications and then developed a MARL-based resource allocation scheme to dynamically adjust the network resources online. For the improvement of long-term QoE performance in dynamic network conditions, the proposed reward function is able to converge efficiently. In addition to this, simulation outcomes showed two significant results: First, for all network traffic loads, semantic communication performs better than conventional communications. Second, with the help of the dynamic reward strategy, the proposed MARL outperforms conventional resource allocation schemes. Furthermore, by incorporating context-aware decision-making, the system adapts to varying environmental and user-specific conditions, enhancing responsiveness. This allows for more intelligent traffic steering, which enables better distribution of network load and optimized use of available resources.

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